

Rachel Hoffman-Bice

CONTACT INFORMATION

d'Arbeloff Laboratory for Information Systems and Technology
77 Massachusetts Ave., Rm 1-007
Cambridge, MA 02139

Cell: (850) 418-3077

Email: rachelmh@mit.edu

Website: <https://www.rachelmhoffman.com>

LinkedIn: www.linkedin.com/in/rachelmhoffmanbice

RESEARCH INTERESTS

- Development of technologies for research in Obstetrics and Gynecology
- Industrial Robotics for Surgical Applications
- Development of technologies for research in Obstetrics and Gynecology
- Deterministic Design of Complex Medical Tools
- Automating Assembly Tasks Through Cable-Driven Robotic Systems
- Development of Robotic Systems for Factory Automation

EDUCATION

Massachusetts Institute of Technology (MIT), Cambridge, MA, USA

Ph.D. Mechanical Engineering, GPA: 4.9/5.0

Expected June 2021

Minor: Leadership and Management

Advisor: H. Harry Asada, Ford Professor of Engineering

Committee: H. Harry Asada, David Trumper, Alberto Rodriguez

S.M. Mechanical Engineering, GPA: 4.9/5.0

June 2017

Thesis: "A Multi-Track Elevator System for E-commerce Fulfillment Centers "

Advisor: H. Harry Asada, Ford Professor of Engineering

University of Florida (UF), Gainesville, FL, USA

B.S. Mechanical Engineering, GPA: 3.97/4.0

June 2015

Minor: Electrical Engineering

Thesis: "A Method for Obtaining Formula One Race Track Curvature for use in Solving Minimum Lap Time Optimal Control Problems"

Advisor: Anil V. Rao, Erich Farber Faculty Fellow

HONORS AND AWARDS

Winner, Best Application Paper Award, IEEE IROS

2017

"A Multi-Track Elevator System for E-Commerce Fulfillment Centers"

University of Florida, Graduated *summa cum laude*

2015

Winner, University of Florida, Outstanding Scholar Commencement Award

2015

Winner, University of Florida, Fyfe Family Foundation Scholarship

2014

Winner, University of Florida, Pratt Whitney Scholarship

2013

RESEARCH EXPERIENCE

Massachusetts Institute of Technology (MIT), Cambridge, MA, USA

Ph.D. Research

2019 - Present

"Using Under-actuated Cable-Driven Robotic Systems for Precise Assembly of Heavy Parts"

d'Arbeloff Laboratory for Information Science and Systems, H. Harry Asada

- Investigated methods by which heavy shafts suspended with multiple cables at specific positions and orientations can be inserted into a chamfered hole despite a small clearance
- Analyzed key conditions for successful peg insertion by analyzing different cable attachment positions and orientations
- Developing a scaled prototype of a three-winch system with linear actuators designed to manipulate a 10 kg payload, including software and control of the system

Temporary Ph.D Research

2018-2019

d'Arbeloff Laboratory for Information Science and Systems, H. Harry Asada

- Developed a novel robotic solution to collaborate with workers to install wire harnesses
- Main research tasks include developing a control strategy coordinating a dual-arm robotic system for positioning wire harness while avoiding collisions
- Implemented Dual-Arm Impedance and Admittance Control for safe human-robot interaction

Master's Research

2015 - 2017

"A Multi-Track Elevator System for E-Commerce Fulfillment Centers"

d'Arbeloff Laboratory for Information Science and Systems, H. Harry Asada

- Invented a novel mechanism by which elevators can travel along tracks, and transfer between track segments of varying direction. This mechanism was inspired by a railway turnout and the rack-pinion drive system of cog-railways.
- Analyzed optimal scheduling routes for multiple carts using this system and detailed the advantages over current elevator systems in e-commerce fulfillment centers.
- Designed and manufactured/implemented scale model consisting of thirteen direction changing mechanisms and twenty-seven track segments

University of Florida (UF), Gainesville, FL, USA

Undergraduate Thesis

2014 - 2015

"A Method for Obtaining Formula One Race Track Curvature for use in Solving Minimum Lap Time Optimal Control Problems"

Vehicle Dynamics and Optimization Laboratory, Anil V. Rao

- Developed user-interactive MATLAB image processing software to obtain formula one track curvature and parametric equations for use in optimal control problems solved in GPOPS-II software
- Allows for users to upload an image of a desired formula one track for analysing minimal lap time

TEACHING
EXPERIENCE

Massachusetts Institute of Technology, Cambridge, MA, USA

Laboratory Teaching Assistant

Spring 2020

2.12/2.120: Introduction to Robotics, H. H. Asada, K. Youcef-Toumi, H. Chin

- Led TAs and Undergrad LAs in providing instruction during laboratory teaching sessions.
- Consulted and mentored two student-teams on the design of a three degree of freedom robotic arm specific to the final project challenge of sorting and throwing plastic bottles to be recycled.
- Provided mentorship to the students through teaching them robot mechanism design and machine design fundamentals and good practices
- Aided the instructors in administration, grading, and laboratory instruction
- Implemented Robotic MATLAB simulations, involving kinematics, trajectory planning and object manipulation, for students continuing education at home due to COVID-19

PROFESSIONAL
EXPERIENCE

MMID, Delft, Netherlands

Functionality Design Intern

Summer 2017

Extensive concept development of a high-pressure pipe coupling, an apparatus for testing chemical dilution and a versatile electric outlet wall box, through use of verbal and non-verbal communication, including detailed hand-sketched concepts. Performed mechanical tests on wall mounted clamp prototypes to ensure material selection and design analysis met client specifications

MIT Lincoln Laboratory, Lincoln, MA, USA

Mechanical Engineering Intern, Systems and Analysis Group

Summer 2014

Evaluated and tested strategies to develop optimal terrain following aircraft trajectory method. Designed and integrated terrain following MATLAB algorithm into radar testing software. Tested and analyzed results from improved radar testing software to ensure problem resolution

Cummins Inc., Columbus, IN, USA

Product Validation Intern

Summer 2013

Analyzed the failure of a fuel filter and the implementation of preventative measures to ensure failure does not reoccur. Applied investigative problem solving methods to determine sources of failure for high pressure fuel lines, including resonant frequency and surface finish analysis

**MENTORSHIP AND
SERVICE**

Baker House, Cambridge, MA, USA

Graduate Resident Advisor

2016 - Present

Foster an inclusive, respectful, supportive, safe, and positive living environment through mentoring, educating, encouraging personal growth, providing outlets for managing stress, and facilitating positive interpersonal relationships. I also Coordinated the Resident Peer Mentor within an undergraduate residence hall and organized and structured inaugural group of students selected to be departmental peer advisors.

MIT Medical Device Design Course, Cambridge, MA, USA

Mentor

2017 - 2019

Provide extensive mechanical design consulting for student-lead medical device design teams, including the development of a Crank-Powered Otoscope, a Center-of-Gravity Detector for Heavy Machinery Operator Fatigue, a Lower Extremity Fracture Fixation Device, a Preterm Vaginal Breech Delivery Device and a Portable Device for helping the physically impaired stand.

UF Peer Advisors Program, Gainesville, FL, USA

Student Organizer, Mentor

2014-2015

Coordinated with faculty to introduce the Peer Advising and Peer Mentor programs into the Mechanical and Aerospace Engineering Department. I was responsible for guiding students through the course selection process and advising them about clubs, available resources, research opportunities and suggested course loads.

MIT Women's Initiative, Cambridge, MA, USA

Participant

January 2018

Traveled to Detroit, MI to speak at 10 different public elementary and middle schools about the value of women in STEM

Pi Tau Sigma, University of Florida Chapter, Gainesville, FL, USA

Member

2010-2012

Coordinated Undergraduate Research Symposium

**OTHER PROJECTS
AND CAMPUS
INVOLVEMENT**

Design of a Disinfection Robot, Cambridge, MA, USA

d'Arbeloff Laboratory for Information Systems and Technology

2020

Design and Fabrication of a mobile platform capable of driving around areas on campus and using

high-powered UV disinfection lights to cleans areas periodically to prevent the spread of Covid-19

Design of a T-Based Lathe, Cambridge, MA, USA

MIT Precision Machine Design Course

2018

Deterministic design, fabrication and testing of a custom T-Based Lathe. The parts cur on the lathe measured flatness 187.9 um +/- 5.75 um and the Error Budget Predicted flatness 233 um +/-22.7um

Control System for Recreating realistic human reaching movements in a direct drive robot arm, Cambridge, MA, USA

Biomechanics and Neural Control of Movement Course

2016

Development of a 2DOF direct drive robot arm with control system representing joint-level reflex feedback, musculoskeletal simulation and the translation of those signals into torque values commanded to the robot.

Design of a Foldable Electric Scooter, Gainesville, FL, USA

Capstone Project

2018

Design and fabrication of a foldable electric scooter capable of carrying a 220 lb adult. The scooter can reach 15.5 mph top speed with a 1.5 s stop time.

DrumBot, Gainesville, FL, USA

Intelligent Robots and Machines Lab

2015

Design,fabrication and programming of a mobile platform with a miniature drum kit attached. The motions of the user are captured by an XBOX Kinect and translated into motions of drumsticks attached to the drumkit.

Drumline, Cymbals, Gainesville FL, USA

UF Marching Band

2010-2012

Performed at the 2012 Olympics in a parade along the Mall in front of Buckingham Palace

COMPUTER SKILLS Software and Frameworks: ROS, OpenCV,Arduino, Gitlab, Github
Languages: Python, MATLAB, L^AT_EX
Applications: SolidWorks, HSMWorks, Microsoft Office Suite/Teams/SharePoint, Slack, VSDC Video Editor

CITIZENSHIP United States of America

REFERENCES Available upon request.

JOURNAL
PUBLICATIONS **R. Hoffman** and H. H. Asada, "Precision Assembly of Heavy Objects Suspended with Multiple Cables From a Crane," *IEEE Robotics and Automation Letters (RA-L)*, Accepted June 2020. *The contents of this paper were also selected by the IEEE International Conference on Intelligent Robots and Systems (IROS) 2020 Program Committee for presentation at the Conference in Las Vegas, Nevada, USA.*

J. Guggenheim, **R. Hoffman**, H. Song and H. H. Asada, "Leveraging the Human Operator in the Design and Control of Supernumerary Robotic Limbs," *IEEE Robotics and Automation Letters (RA-L)*, vol. 5, no. 2, pp. 2177-2184, April 2020. *The contents of this paper were also selected by the IEEE International Conference of Robotics and Automation (ICRA) 2020 Program Committee for presentation at the Conference in Paris, France.*

CONFERENCE
PUBLICATIONS

R. Hoffman, and H.H. Asada, “A Multi-Track Elevator System for E-Commerce Fulfillment Centers,” *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2017)*, Vancouver, Canada, October 2017.

J. Chacon-Castano, D. Rathone, **R. Hoffman**, H. Yang, D. Pantazis, J. Yang and N. Hanumara, “Music and the Brain – Design of an MEG Compatible Piano,” *2017 IEEE International International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2017)*, Seogwipo, South Korea, July 2017.